greendash

The green recommendation that shows its receipts · M1: Approach · Green Customer Insights Platform (Lloyds)

Team: Kratik, Akshit, Aniket, Madhav, Ritvik · 10 October 2026

1. Problem statement and our understanding

Lloyds asks for an AI platform that analyses customer transactions, financial products and behaviour to find opportunities for green products (EV Finance, Green Home, Solar Finance and Paperless Banking), estimate their sustainability impact, explain each one to customers, and give business teams actionable insight. The brief sets a strict rule: **ML predicts, a rules-based carbon engine calculates, and the LLM only explains**; financial eligibility comes from product rules, never from a model.

Our reading: the hard part is not measuring carbon, it is **trust and actionability**. Customers want greener choices but cannot tell which options fit them, what they would really save, or whether the bank's claim is credible. Banks have green products and green-intent customers but no **auditable** bridge between them. Today's carbon trackers stop at “here is your footprint”, blend likelihood, eligibility and impact into one opaque score, show one confident number built on coarse averages, and treat missing data as a failure. In green finance, a confident AI that cannot show its working is a greenwashing risk. The brief's evaluation reflects this: it tests missing data, conflicting signals, products that must not be recommended and insufficient information to estimate impact.

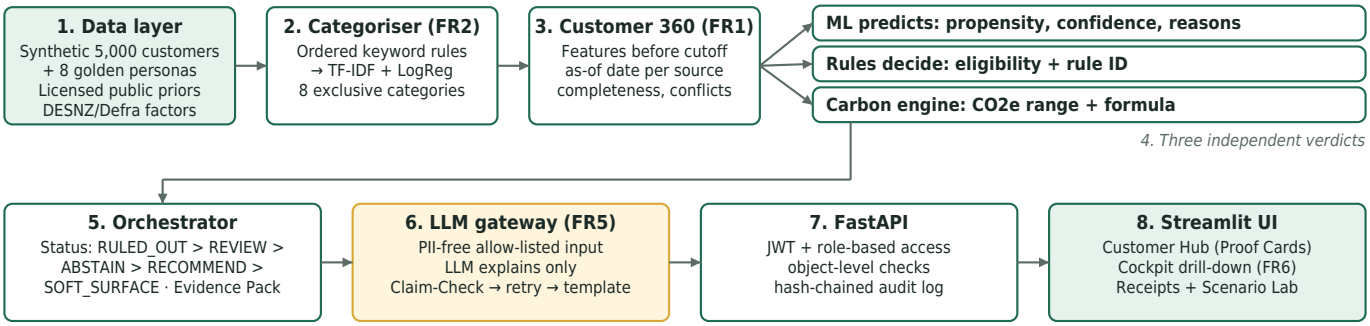
2. Idea / abstract

greendash turns everyday banking data into personalised green next steps for customers and an evidence-backed opportunity pipeline for the bank. Every insight is a **Proof Card**: a recommendation that carries its own receipt.

- **Three separate verdicts that never merge:** Likelihood (calibrated ML propensity), Eligibility (versioned product rules, with the rule ID that fired) and Impact (a low–high CO_2e range from official DESNZ/Defra factors, with formula, baseline and horizon, labelled “Estimate, not a guarantee”).
- **Evidence with dates and decomposed confidence** (data completeness, recency, model stability, in-distribution), so customers see why the card appeared and how sure we are.
- **Refusals are designed states:** RULED_OUT explains “why not”, ABSTAIN asks for the one missing field, and REVIEW flags mixed signals, so the system never invents a number.
- **Claim-Check firewall:** a deterministic verifier checks every LLM sentence against the engine outputs (no new numbers, no “guaranteed / will save / pre-approved”) before anyone sees it; failures fall back to a safe template.
- **Scenario Lab:** judges mutate a customer live (set vehicle to EV, remove mileage, add a conflict, make data stale) and watch all three verdicts recompute with the exact cause.

3. High-level approach, pipeline and architecture

One Python 3.11 codebase of pure-function engines behind typed Pydantic contracts, reproducible end to end with **make all** (fixed seed, hash manifest). It follows the brief's pipeline: Data → Feature store → Categorisation, propensity models, carbon engine → Recommendation engine → LLM insight layer → Dashboard and API.



Component	Approach
Data (synthetic + legal)	No real customer data. Seeded generator: 5,000 customers + 8 golden personas across the brief's 8 tables. Hidden drivers (green affinity, financial headroom, life events, noise) shape behaviour. Licence-gated public data (OGL / CC BY, e.g. DfT vehicle and mileage statistics, Ofgem consumption values) is used only as distribution priors. Cutoff 30 Jun 2026; take-up = adoption within a **90-day** window.
Labels	Take-up = Bernoulli(sigmoid(observed signals + hidden drivers + noise)), so labels are plausible but not trivially separable. Category labels come from the merchant template; all rules documented in DATA.md.
Categorisation (FR2)	Ordered keyword rules, then TF-IDF character n-gram logistic regression; documented boundary decisions (public EV charging = Fuel, dual-fuel bill = Utilities, taxis = Other).
Propensity (FR3)	One LightGBM model per product, isotonic-calibrated, monotone constraints; SHAP grouped into fixed reason codes (e.g. HIGH_FUEL_SPEND). Baselines: logistic regression and a rule heuristic.
Carbon engine (FR4)	Deterministic per-product formulas (e.g. petrol miles × factor − EV miles × kWh/mile × grid factor), pinned factor vintage traceable to sheet and row, avoided-emissions framing per GHG Protocol guidance. Paperless reports paper sheets and digital adoption only, never CO2e.
Recommendation	Orchestrator applies status rules in order (rules override the model) and builds an Evidence Pack; it never edits a verdict.

Component	Approach
LLM + security	Small model at temperature 0, cached. It sees only allow-listed, engine-formatted strings: no raw transactions, names, postcodes or free text, which blocks prompt injection at the input. PII scan before and after the call; JWT + role-based access on the API; encrypted sensitive fields; tamper-evident audit log.
Product (FR6, FR7)	Streamlit: Customer Hub (mobile-first Proof Cards), Cockpit (CO2e, energy, paper and digital adoption reported separately; funnel; propensity × impact quadrant; drill-down Customer → Opportunity → Evidence → Impact → Action) and a Receipts page showing every metric.

4. Planned testing methodology

Every claim we make is backed by an automated test or a metric generated from one command into the evaluation report (on the validation set; final numbers on the held-out test set).

Area	Method	Metrics / pass criteria
Data quality & leakage	Customer-level 70/15/15 split; schema checks (pandera); automated leakage tests (no row after cutoff, no customer in two splits); placebo test with shuffled labels	0 leaks; placebo AUC ≈ 0.5; base rates in target ranges; no single feature AUC above 0.80; 8/8 personas match designed outcomes
Propensity models	Compare LightGBM with logistic regression and rule-heuristic baselines on the same splits	ROC-AUC, PR-AUC, Brier score, calibration curve and error, precision and lift at top 10% vs both baselines
Confidence	Group validation predictions into confidence terciles	Brier score falls from low to high tercile; low correlation with propensity
Categorisation	Hold out whole merchants the model never saw; 200-string hand-checked gold set	Macro-F1 vs rules-only baseline, confusion matrix, worked examples
Carbon engine	Hand-calculated unit test for every formula; factor values checked against the source workbook	Exact match to hand calculations; INSUFFICIENT_DATA when inputs are missing
LLM explanations	150-case evaluation set: 8 personas × perturbations (missing, conflicting, ruled-out, abstain, stale) + 20 adversarial sentences + prompt-injection cases; governed pipeline vs a naive prompt	Numeric fidelity, unsupported-claim rate, banned-phrase rate, product-mapping accuracy, correct abstain wording, fallback rate, latency, cost per 1,000 insights
Organiser scenarios	30+ YAML scenario cases run headlessly through the Scenario Lab engine; invariance test (Lab output equals a fresh run on mutated data)	Pass/fail per case across missing data, conflicting signals, ruled-out products and insufficient impact data
Security	Tests for cross-customer access (IDOR), role limits, PII reaching the LLM, injected instructions, audit tampering, SQL injection; secret scanning and dependency audit in CI	All blocked; red-team claim blocked with exactly the expected violations
System	Clean-clone rerun; latency benchmark	Identical artefact hashes; scenario recompute under 1 s

Limitations we state openly: labels are synthetic, so metrics are an upper bound and show the pipeline is honest, not real-world accuracy. All impact figures are modelled estimates, and nothing in greendash is a credit decision.